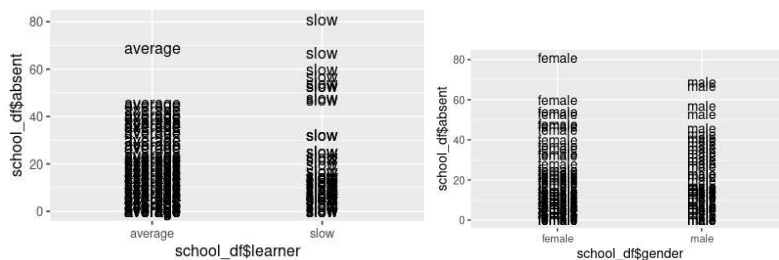


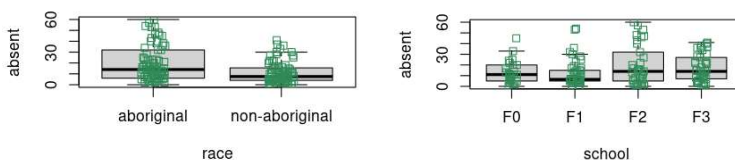
Executive Summary: Case Study 2

Our objective in conducting this case study was to determine whether cultural origin, school type, type of learner, gender, and interactions were related to the amount of days that a child was absent from school. Before we could begin conducting our analysis, we had to clean the data. We found that there was a redundant column in our dataset, so we removed it. We also replaced any categorical variables with factor variables. There were 4 categorical variables: race, gender, school level, and learner type, so we used the `as.factor` function to replace them all.

Once we cleaned the data, we began fitting our ANOVA model. First, we tested our predictors. We were first able to eliminate Learner because it was too random (learner = 0 and learner = 1) were pretty much the same. Then, we tested gender. We decided to eliminate gender for the same reasons.

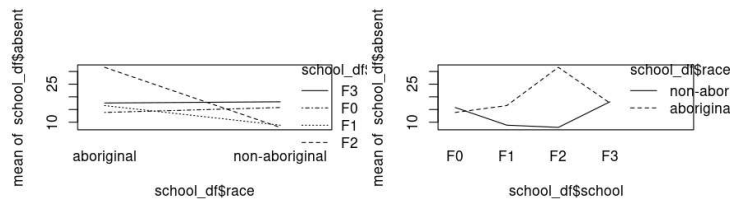


For the race and the school level predictors, we decided to use box plots. For this, we reverted back to our original dataset, with the categorical variables. When we looked at both predictors, we could not certainly determine whether the observed differences were statistically significant.



Since we couldn't certainly tell whether or not the race and school terms were statistically significant, we created interaction plots for the two predictors, one with school plotted against absent, and

one with race plotted against absent. Only the one with race plotted against school matters, because that has the intersecting lines at the edge.



We then started to create a model. The research question that we wanted to investigate was whether a multiplicative model or an additive model for race and school would better model their relationship to absences. School.full wouldn't have a race and gender interaction term, because there is no interaction between the two variables, as we can see from the interaction plot. We chose to use $\alpha = .1$ as our threshold for statistically significant interaction term. We created an Anova table, and from that, we found that the race: school term is statistically significant, as well as school, and race. The next thing we did was create a confidence interval. Our null hypothesis was that there would be no difference in mean for any of the factors, and our alternative hypothesis was that there would be a mean difference in at least one of the factors.

After that, we selected our model. We tested three different combinations, and the one that we found was the best was with race + school as the predictor. This was because the interaction term was shown to be insignificant, and school was shown to be a little off significant, so the final model had race + school. The next thing we did was create separate confidence intervals of race and school. We found that the CI of school was more randomly distributed than the CI of race, which is why we used race as the predictor of absence.

The next thing we did was check for any unusual observations. There were no high leverage points or highly influential points, but there were 10 outliers. After that, we checked our model assumptions, constant variance and normality. We used a Breusch-Pagan test to check for constant variance, and with that we were able to determine that the p-value was greater than the alpha of .05, so we failed to reject the

null and conclude that our variance was constant. We conducted a Kolgorov-Smirsch test to check for normality, and we found that our p-value was less than our alpha of .05, so we rejected the null and concluded that our model was not normal.

Because our model wasn't normal, we tried to use a box-cox transformation. However, we found that it wasn't possible, because there were too many data points with an absence of 0. We then tested several different ways to move the absence column to allow for all absences to be positive, but when it showed that it still was not normal, we had to conclude that the error could not be normalized.

In conclusion, we found that the best model for the relation between race, school and absence was the additive one. This means that race and school level being independent is the better way to potentially predict the number of absences a student would have, rather than a multiplicative model, which would have them dependent on one another.